# MARK SCHEME · TEACHER COPY

10BR · Chapter 4: Software

## Cambridge IGCSE Computer Science 0478

Paper 1  Computer Systems

## Total marks: 75

Official Cambridge mark scheme extracts for each question in the matching exam paper, with question numbers changed to match the exam. Source of each question is shown above its rows.

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## Question 1 · May/June 2023 Paper 11, original Q4 [4 marks]

| Question | Answer | Marks |
|---|---|---|
| 1 | **One** mark for each correct term in the correct place:<br>• System<br>• Application<br>• Operating<br>• Hardware | 4 |

## Question 2 · May/June 2023 Paper 12, original Q10 (a),(b) [5 marks]

| Question | Answer | Marks |
|---|---|---|
| 2(a) | **Two** from:<br>• System software provides services that the computer requires<br>• … whereas application software provides services that the user requires<br><br>**One** from (system software):<br>• Utility software // by example e.g. defragmentation software, antivirus, firewall<br>• Operating system<br><br>**One** from (application software):<br>• Any suitable example of an application e.g. word processor, web browser, video-editing software | 4 |
| 2(b) | • Secondary storage // HDD // SSD | 1 |

## Question 3 · May/June 2023 Paper 13, original Q4 (a),(b) [5 marks]

| Question | Answer | Marks |
|---|---|---|
| 3(a) | • B | 1 |
| 3(b)(i) | • Machine code // low-level language // object code | 1 |
| 3(b)(ii) | • Interpreter | 1 |
| 3(b)(iii) | • Compiler | 1 |
| 3(b)(iv) | • Compiler | 1 |

## Question 4 · Oct/Nov 2023 Paper 11, original Q9 (a),(b),(c) [10 marks]

| Question | Answer | Marks |
|---|---|---|
| 4(a) | Any **one** from:<br>– Operating system // Interrupt handler | 1 |
| 4(b) | Any **five** from:<br>e.g.<br>– Key press generates the interrupt<br>– Interrupt given a priority<br>– Interrupt is sent to CPU<br>– Interrupt is placed in a queue<br>– CPU stops current task to check the queue/service the interrupt …<br>– … using an interrupt service routine<br>– If key press is highest priority the interrupt is processed | 5 |
| 4(c)(i) | Any **two** suitable hardware example e.g.:<br>– Moving the mouse<br>– Clicking a mouse button<br>– Plugging in a device<br>– Paper jam in printer<br>– Printer out of paper | 2 |
| 4(c)(ii) | Any **two** suitable software examples e.g.:<br>– Division by zero<br>– Two processes accessing the same memory location<br>– Null value | 2 |

## Question 5 · May/June 2024 Paper 12, original Q4 (a),(b),(c) [5 marks]

| Question | Answer | Marks |
|---|---|---|
| 5(a) | Any **one** from:<br>• Operating system<br>• Utility software // by example | 1 |

**Total: 75 marks**

| | | |
|---|---|---|
| 5(b) | Any **two** from.:<br>Example:<br>• Word processor<br>• Spreadsheet<br>• Database<br>• Web browser<br>• Image/photo-editor<br>• Video-editor<br>• Email client<br>• Game<br>• Social media app | **2** |
| 5(c) | Any **two** from:<br>• System software **manages/maintains** the **hardware/software**<br>• Applications software allows the **user** to perform tasks | **2** |

## Question 6 · Oct/Nov 2024 Paper 11, original Q7 (a),(b) [4 marks]

| Question | Answer | Marks |
|---|---|---|
| 6(a)(i) i) | D | 1 |

**Total: 75 marks**

| | | |
|---|---|---|
| 6(a)(ii) | **One** mark for identifying a function. **One** mark for a matching description.<br>For example:<br><br>• Memory management<br>• … managing what gets allocated where in memory<br><br>• Managing peripherals and drivers<br>• … managing the communication between any input and output devices that are connected to the computer<br><br>• Multitasking<br>• … managing the process of **switching** between tasks that are being carrying out<br><br>• Platform for running applications<br>• … allowing communication between the applications software and the hardware<br><br>• System security<br>• … proving features such as username and password<br><br>• User accounts<br>• … allowing multiple user accounts to be created on a computer | **2** |
| 6(b) | Any **one** from:<br><br>• Firmware<br>• Bootloader<br>• BIOS<br>• Bootstrap | **1** |

## Question 7 · Oct/Nov 2024 Paper 12, original Q1 (a),(b),(c) [3 marks]

| Question | Answer | Marks |
|---|---|---|
| 7(a) i) | **Instructions/program** that is used to operate a computer/hardware | 1 |
| 7(b) | B | 1 |
| 7(c) | Operating system // system software | 1 |

**Question 8 · Feb/March 2025 Paper 12, original Q7 (a),(b),(c) [9 marks]**

| Question | Answer | Marks |
|---|---|---|
| 8(a) | • Language: assembly<br>• Translator: assembler | 2 |
| 8(b) | Any four from<br>• Each statement is translated and executed before the next<br>• **Translator stops** when an error is found<br>• … can correct and continue from where error found // **real time** debugging when/where error found<br>• ... does not need to retranslate all the code again<br>• Can test sections of the code<br>• Can test without all the program code being functional/accurate/complete | 4 |
| 8(c) | Any three from:<br>Example<br>• Code editor<br>• Run-time environment<br>• Error diagnostics // by example<br>• Auto-complete<br>• Auto-correct<br>• Prettyprint // syntax highlighting<br>• Collapse/expand block<br>• Auto-**documentation** | 3 |

**Total: 75 marks**

## Question 9 · May/June 2025 Paper 13, original Q1 (b) [1 marks]

| Question | Answer | Marks |
|---|---|---|
| 9(a) | A | 1 |

## Question 10 · May/June 2025 Paper 13, original Q4 (a),(b),(c) [6 marks]

| Question | Answer | Marks |
|---|---|---|
| 10(a) | Any **two** from:<br>• It assigns a priority to an interrupt.<br>• It uses an interrupt handler / interrupt service routine to process the interrupt.<br>• Maintains/organises the interrupt queue | 2 |
| 10(b) | Any **two** from:<br>Examples:<br>• managing files<br>• providing an interface<br>• managing peripherals and drivers<br>• managing memory<br>• managing multitasking<br>• providing a platform for running applications<br>• providing system security<br>• managing user accounts | 2 |
| 10(c)(i) | Any **one** from:<br>Examples:<br>• key press on a keyboard<br>• button clicks on a mouse<br>• printer out of ink | 1 |

**Total: 75 marks**

| | | |
|---|---|---|
| 10(c)(ii) | Any **one** from:<br><br>Examples:<br><br>• division by zero<br>• two processes trying to access the same memory location | **1** |

## Question 11 · Oct/Nov 2025 Paper 12, original Q3 (a),(b),(c) [4 marks]

| Question | Answer | Marks |
|---|---|---|
| 11(a) | Any **two** from:<br><br>• Low-level language<br>• Uses mnemonics<br>• Used to **communicate directly** with the computer hardware | **2** |
| 11(b) i | Assembler | **1** |
| 11(c) :) | B | **1** |

## Question 12 · May/June 2026 Paper 11, original Q5 (a),(b),(c),(d) [10 marks]

| Question | Answer | Marks | Guidance |
|---|---|---|---|
| 12 | Examiners must use the **full range of marks** defined in the mark scheme, unless limited by the quality of the candidate responses seen. | | |
| 12(a) | To translate/convert high-level language into low-level language/machine code/binary | **1** | Just translates code/program to low-level/machine code is NE |
| 12(b) | Assembler | **1** | Mark the first answer given.<br><br>Do not accept 'assembly'. |

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|---|---|---|---|
| 12(c) | **One** mark for each correct letter<br>**One** mark for each corrected statement<br><br>• A<br>• An interpreter translates and executes the code **one line/statement at a time**<br><br>• E<br>• A compiler produces an **executable** file | **4** | If an incorrect letter is given, do not read on to statement correction.<br><br>Accept minor variations for correct statements that have the same meaning.<br><br>For letter A we are looking for a correction on the use of variable.<br><br>For letter E we are looking for a correction on the use of non-executable file.<br><br>If no answer is given in answer space, look to see if indication is given in the table. |

| | | | |
|---|---|---|---|
| 12(d) | **One** mark for each correct function<br>**One** mark for a matching purpose<br><br>• Code editor<br>• Allows the **user/programmer** to write/change the program<br><br>• Run-time environment<br>• Allows the user to **run/execute** the code and see the **output/result** // Allows the user to **input data** to **run/execute** the code // To allow the user to test the code<br><br>• Error diagnostics // debugging tools // by example e.g. break points, stepping, variable watch window<br>• Finds/identifies/detects errors in the code // by example<br><br>• Auto-completion<br>• Gives **options** to complete the remainder of a command word/key term/identifier/syntax (to limit errors/speed up development) // It **predicts** the rest of a command word/key term/identifier (to limit errors/speed up development) // It **predicts** the required syntax e.g. when a bracket is needed (to limit errors/speed up development)<br><br>• Auto-correction<br>• Corrects **spelling** errors in command words/key terms/identifiers/syntax (to limit errors in the code)<br><br>• Prettyprint // syntax highlighting // colour coding<br>• Changes the colour of text in the code to highlight command words/key terms/identifiers (to make them easier to identify)<br><br>• Line numbers<br>• To give (incremented) numbers to each line in the code (to allow a line in the code to be easily/quickly found) | **4** | Do not accept translator/interpreter/compiler, this is given in the question.<br><br>Mark first function given.<br><br>If the function name is TV/blank, read on to the purpose to see if a correct function name is included. Award both correct name and purpose if seen. Award the purpose if the function name is still TV but it is obvious what is meant.<br><br>Code editor – BOD just editor alone<br>Code editor purpose – BOD edit in place of write/change<br><br>Error diagnostics – just 'debugging' alone is TV<br><br>Just 'words' alone is TV for command word/key term/identifier/syntax<br><br>Allow minor spelling variations on function names. It you see something that you think is a viable alternate to the function name, please refer to your TL.<br><br>Other viable functions outside the syllabus may be seen, such as:<br><br>• artificial intelligence<br>• auto-documentation<br>• expand/collapse code blocks<br>• built-in libraries/functions<br><br>In this case, accept the function and a correct description of the purpose. The description must be specific and not generic. Refer to TL if in doubt. |

**Total: 75 marks**

## Question 13 · May/June 2026 Paper 12, original Q7 (a),(b),(c),(d),(e) [9 marks]

| Question | Answer | Marks | Guidance |
|---|---|---|---|
| 13(a) | A language that is closer to the language processed/used by a computer<br>// A language that is closer to binary/machine code<br>// A language has that little abstraction from that processed by a computer | 1 | Do not accept 'understood' in place of processed/used.<br><br>'it uses/is binary' is TV for it is closer to binary.<br><br>Do not accept that it is a language that uses mnemonics. This is not answering this question as not all do.<br><br>Do not accept benefits e.g. it can directly manipulate the hardware. This is not answering the question.<br><br>Do not accept marks for saying what it is not e.g. It is not close to human language. |
| 13(b) | **Two** from:<br><br>• Assembly (language)<br>• Machine code // Binary<br>• Object code | 2 | Mark first answer given.<br><br>MP2 – BOD machine language |
| 13(c) | Assembler | 1 | Mark first answer given. |
| 13(d) | **Two** from:<br><br>• The program doesn't need to be portable<br>• Can **directly manipulate/access** the hardware // Can have low-level **control** of the device<br>• The program will be faster to **execute/run**<br>• The program can be written for **specific** hardware // Can use/access **specialised** hardware<br>• The program uses **memory** more **efficiently** // The program is less memory intensive<br>• The program would take up less **storage** space (in the vending machine) // The **file size** would be smaller (than the high-level equivalent file) | 2 | Mark whole answer.<br><br>MP2 – program can communicate with hardware is TV as high-level language can also do this. There is no reference to it being direct.<br><br>MP3 – Less time to translate is NE for execute<br><br>MP5 – uses less space in memory is TV |

**Total: 75 marks**

| | | | |
|---|---|---|---|
| 13(e) | • Auto-correction<br>• Run-time environment<br>• PrettyPrint | **3** | Mark first answer given. |

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